

ENCN445: Environmental Fluid Mechanics

Coastal Engineering Tutorials:

Tutorial 2

Problem One:

A wave is travelling through a 2000 m deep ocean with $\omega = 0.495 \text{ rad/s}$, $k = 0.025 \text{ rad/m}$, and $\eta_0 = 0.3 \text{ m}$.

A Closer to the shore the ocean depth has decreased (gradually) to only 100 m. Calculate the angular frequency, wavenumber, and phase velocity at this new depth.

B Finally, calculate the properties of this wave once it has reached a fluid depth of 4 m. This time calculate the same properties as in B but also calculate the new wavelength, the amplitude and the group velocity of this and the original wave.

Problem Two:

Waves are approaching a beach with a wavelength of 150 m and an amplitude of 0.7 m. The ocean depth is 4 m. The waves are approaching bathymetric contours (lines of constant depth) at an angle of 15 degrees (or $\pi/12$ radians.)

A Calculate the angle that the waves will be at relative to the bathymetric contours when the ocean is 3 m deep.

B Calculate the wave amplitude when the ocean is 3 m deep.

C What are the refraction and shoaling coefficients for the above situation?

D There is a small jetty at the beach. Local users want to protect the jetty with a breakwater placed where the water depth is 3 m. There is concern about the interaction of reflected waves with the original wavefield for the safety of small boats however. Two designs have been proposed: The first is a concrete seawall with an estimated reflection coefficient of 0.7 and the second is a rock breakwater with a reflection coefficient of 0.5. The concrete seawall would be preferred by the local group as it is cheaper. If the reflected waves need to have an amplitude less than 0.5 m for safety reasons which structure should be built. You can assume that the wave conditions above are typical.

E Comment on how long the breakwater should be in relation to the area that needs to be protected. Are there any ways to reduce the length of the breakwater without impacting on its efficacy?

Problem Three:

A keen surfer and fluid dynamicist is visiting a new area and wanting to know which local beach is most likely to have the best waves. From a fluid dynamics point of view, this corresponds to plunging waves rather than spilling or surging waves. Whereas most people would just use the internet or ask locals, this particular surfer wants to use their knowledge of fluid dynamics to determine where to go surfing, thus satisfying both of their passions. There are three local beaches with bed slopes of 0.8° , 1.6° , and 4.5° respectively. The waves that are approaching this section of beach have a deep water wavelength of 200 m and a deep water wave height of 1 m.

A Determine which beach is most likely to have plunging waves based on the above characteristics

B Calculate how far out the surfer will have to paddle to get to the location where the waves are breaking. Assume that the waves are approaching the beach directly (no refraction), the beach slope is constant, our wave solution is valid where the waves are breaking, and a value for the breaking index κ of 1.15. You can also assume that where the wave is breaking it is a shallow water wave.